

C8.4 Probabilistic Combinatorics: Sheet 4

Components in $G(n, p)$, Janson's inequalities

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Problem 1

Let X and Y be independent with $X \sim \text{Po}(c)$ and $Y \sim \text{Po}(d)$. Show that $X + Y \sim \text{Po}(c + d)$. Show that the conditional distribution of X , given that $X + Y = n$, is binomial with parameters n and $c/(c + d)$. Deduce that if $Z \sim \text{Po}(a)$ and the conditional distribution of W given that $Z = n$ is $\text{Bin}(n, p)$, then $W \sim \text{Po}(ap)$, $Z - W \sim \text{Po}(a(1 - p))$ and W and $Z - W$ are independent.

Proof.

□

Problem 2

Consider $G(n, p)$ with $p \sim \frac{\log n}{n}$, and let Y_r be the number of components with r vertices.

(a) Using Cayley's formula, show that

$$\mathbb{E}[Y_r] \leq \binom{n}{r} r^{r-2} p^{r-1} (1-p)^{r(n-r)}.$$

Proof. For any set of vertices $S \subseteq V$ with $|S| = r$, any component on S contains a tree that spans S , and there are no edges between S and $V \setminus S$. The equation follow from Cayley's formula. $\square$

(b) Hence show that

$$\mathbb{E} \left[\sum_{r=2}^{n/2} Y_r \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Proof. Note that

$$\mathbb{E}[Y_r] \leq \frac{(ne)^r}{r^2} \cdot p^{r-1} e^{-pr(n-r)} \leq r^{-2} p^{-1} (enpe^{-p(n-r)})^r \leq r^{-2} p^{-1} (enpe^{-pn/2})^r$$

Put $x_n = enpe^{-pn/2}$. For large enough n ,

$$x_n \leq 2e(\log n)e^{-0.49 \log n} = o(n^{-1/3}).$$

Thus,

$$\mathbb{E} \left[\sum_{r=2}^{n/2} Y_r \right] \leq \sum_{r=2}^{n/2} \mathbb{E}[Y_r] \leq \sum_{r=2}^{n/2} r^{-2} p^{-1} x_n^r \leq \frac{4}{n} \sum_{r=2}^{n/2} x_n^r = \frac{4x_n^2}{n(1-x_n)} = o(1).$$

$\square$

(c) Deduce that as $n \rightarrow \infty$,

$$\mathbb{P}(\text{all non-isolated vertices are in a single component}) \rightarrow 1$$

for any fixed c .

Proof. By (b), all non-isolated vertices are in a component of size greater than $n/2$. But then there cannot be multiple of such components. $\square$

(d) Deduce that $p(n) = \frac{\log n}{n}$ is a threshold function for $G(n, p)$ to be connected.

Proof. Suppose $\frac{p}{(\log n)/n}$. By Sheet 1 Q8, there are almost surely no isolated vertices, so by part (c) G is almost surely connected. The converse follows the same argument. $\square$

Hints: (a) connected graphs contain at least one spanning tree. (b) it may help to bound $\mathbb{E}[Y_2]$ and $\mathbb{E}[Y_r]$, $3 \leq r \leq n/2$, differently. (d) recall the result of Sheet 1 Q7.

Problem 3

Give an example of three events A , B_1 and B_2 such that $\mathbb{P}(A \mid B_i) > \mathbb{P}(A)$ for $i = 1, 2$ but $\mathbb{P}(A \mid B_1 \cup B_2) < \mathbb{P}(A)$.

Proof. Consider the probability space $\{w, x, y, z\}$ with $p_w = 0.1, p_x = p_y = p_z = 0.3$. Let $A = \{w \cup z\}$, $B_1 = \{w \cup x\}$, $B_2 = \{w \cup y\}$. $\square$

Problem 4

Let $U_1, U_2, \dots$ be up-sets (increasing events) in $\mathcal{P}(X)$, let D be a down-set, and let $\mathbb{P}_p$ denote the probability measure associated to selecting each element of X independently with probability p .

(a) Show that $\mathbb{P}_p(U_1 \cap \dots \cap U_k) \geq \prod_{i=1}^k \mathbb{P}_p(U_i)$.

Proof. This follows immediately from induction and Harris Lemma. □

(b) Show that $\mathbb{P}_p(U_1 \mid D) \leq \mathbb{P}_p(U_1)$.

Proof. By Harris' Lemma,

$$\mathbb{P}_p(U_1 \cap D) = \mathbb{P}_p(U_1) - \mathbb{P}_p(U_1 \cap D^c) \leq \mathbb{P}_p(U_1) - \mathbb{P}_p(U_1)\mathbb{P}_p(D^c) = \mathbb{P}_p(U_1)\mathbb{P}_p(D). \quad \square$$

(c) Show that if U_1 and U_2 are independent, then $\mathbb{P}_p(U_1 \mid U_2 \cap D) \leq \mathbb{P}_p(U_1 \mid U_2)$.

Proof. By Harris Lemma,

$$\begin{aligned} \mathbb{P}_p(U_1 \mid U_2 \cap D) &= \frac{\mathbb{P}_p(U_1 \cap U_2 \cap D)}{\mathbb{P}_p(U_2 \cap D)} \\ &= \frac{\mathbb{P}_p(U_1 \cap U_2) - \mathbb{P}_p(U_1 \cap U_2 \cap D^c)}{\mathbb{P}_p(U_2 \cap D)} \\ &\leq \frac{\mathbb{P}_p(U_1)\mathbb{P}_p(U_2) - \mathbb{P}_p(U_1)\mathbb{P}_p(U_2)\mathbb{P}_p(D^c)}{\mathbb{P}_p(U_2 \cap D)} \\ &= \frac{\mathbb{P}_p(U_1)\mathbb{P}_p(U_2 \cap D)}{\mathbb{P}_p(U_2 \cap D)} = \mathbb{P}_p(U_1). \end{aligned}$$

□

(d) Does the last inequality always hold, i.e., without assuming independence?

Proof. □

Problem 5

Let $U \subseteq \mathcal{P}(X)$ be an up-set. Show that the function $f(p)$ defined by $f(p) := \mathbb{P}_p(U)$ is increasing on $[0, 1]$. Is it continuous? Differentiable?

Proof. Let $0 \leq p \leq q \leq 1$. For $x \in X$, let U_x be a uniform random variable on $[0, 1]$. Let $X_p = \sum_{x \in X} \mathbb{1}_{U_x \leq p} \cdot x$ and $X_q = \sum_{x \in X} \mathbb{1}_{U_x \leq q} \cdot x$. Then $X_p \subseteq X_q$. Thus, since U is an upset, $X_p \in U$ implies $X_q \in U$. It follows that $f(p) \leq f(q)$.

Since $f(p) = \sum_{A \in U} p^{|A|} (1-p)^{n-|A|}$, $f(p)$ is a polynomial, which is continuous and differentiable. $\square$

Problem 6

- (a) When does equality hold in Harris's Lemma?

Proof. We call $\mathcal{A} \subseteq \mathcal{P}(n)$ independent of some $x \in [n]$ if

$$\mathbb{P}(X_p \in \mathcal{A}) = \mathbb{P}(X_p \cup \{n\} \in \mathcal{A}) = \mathbb{P}(X_p \setminus \{n\} \in \mathcal{A}).$$

Equality in Harris's Lemma holds if $\mathcal{A}_0 = \mathcal{A}_1$ or $\mathcal{B}_0 = \mathcal{B}_1$. Say $\mathcal{A}_0 = \mathcal{A}_1$. Then $\mathcal{A}$ is independent of n . But then the same applies replacing n with any other coordinates. Thus, there exists $S \sqcup T = [n]$ such that $\mathcal{A}$ only depends on S and $\mathcal{B}$ only depends on T . $\square$

- (b) Deduce that if U is up-set that is (pairwise) independent of the up-sets U_i for $i = 1, \dots, k$, then U is independent of the family $\{U_1, \dots, U_k\}$.

Proof. By (a), U depends on some set $S \subseteq X$ that is disjoint from $\bigcup_{i=1}^k S_i$, where $S_i \subseteq X$ is the set that U_i depends on. Thus, U is independent of the family $\{U_1, \dots, U_k\}$. $\square$

- (c) Deduce that Janson's inequalities also hold for general up-sets.

Proof. The only part of the proof Janson's inequality that assumed the events A_i were principal up-sets was the assumption that if A_i is independent of A_j for $j \not\prec i$, then it is independent of $\bigcap_{j \not\prec i} A_j$. This follows for general up-sets from (b). $\square$

Problem 7

Let $0 < p < 1$ be fixed, and let X denote the number of triangles in $G(n, p)$. Show that there is a constant $c = c(p) > 0$ such that $\mathbb{P}(X = 0) \leq e^{-cn^2}$ for all $n \geq 3$. Is $\mathbb{P}(X = 0)$ at least $e^{-c'n^2}$ for some constant c' ?

Let $G = G(n, p)$, where $0 < p < 1$ is constant. Call a subset V of $V(G)$ triangle-free if the subgraph of G induced by V contains no triangles. Show that there is a constant A such that whp (i.e., with probability tending to 1) G contains no triangle-free subset with more than $A \log n$ vertices.

Do you expect $G(n, 1/2)$ to have a triangle-free subset of size at least $B \log n$ for some constant B ?

Proof. Let $T_1, \dots, T_M$ be all the possible triangles in $G(n, p)$, where $M = \binom{n}{3}$. For each i , let A_i be the event that T_i is present in $G(n, p)$. Then $\mu = \mathbb{E}[X] = \sum_{i=1}^M \mathbb{P}(A_i) = \Theta(n^3 p^3)$ and $\Delta = \sum_i \sum_{i \sim j} \mathbb{P}(A_i \cap A_j) = \Theta(n^4 p^5)$. Since $\Delta > \mu$, by the second Janson's inequality, $\mathbb{P}(X = 0) \leq e^{-\mu^2/2\Delta} \leq e^{-cn^2}$, for some constant $c = c(p) > 0$.

Note that

$$\mathbb{P}(X = 0) \geq \mathbb{P}(e(G(n, p)) = 0) = (1 - p)^{\binom{n}{2}} \geq \exp(-\Theta(n^2)).$$

For any subset $S \subseteq V(G)$ with $|S| > A \log n$, by Janson,

$$\mathbb{P}(G[S] \text{ is triangle-free}) \leq \exp(-c(A \log n)^2) = n^{-cA^2 \log n}.$$

By the union bound,

$$\mathbb{P}\left(\bigcup_{\substack{S \subseteq V(G) \\ |S| > A \log n}} \mathbb{P}(G[S] \text{ is triangle-free})\right) \leq n^{A \log n} \cdot n^{-cA^2 \log n} = \exp((1 - Ac)A(\log n)^2) = o(1),$$

for $A > 1/c$. □

Problem 8

Let H be a fixed strictly balanced graph, let $0 < \alpha < |V(H)|/|E(H)|$ be fixed, let $p = p(n) = n^{-\alpha}$, and let X denote the number of copies of H in $G(n, p)$. Use Janson's inequality to show that there is some constant $\beta > 0$ such that $\mathbb{P}(X = 0) \leq \exp(-n^\beta)$ for n sufficiently large.

Proof. Put $v = |V(H)|$, $e = |E(H)|$. Let $H_1, \dots, H_m$ be all possible copies of H in $G(n, p)$. For each i , let A_i be the event that H_i is present in $G(n, p)$. Then

$$\mu = \mathbb{E}[X] = \sum_i \mathbb{P}(A_i) = \Theta(n^v p^e) = \Theta(n^c),$$

where $c = v - \alpha e > 0$, and $\Delta = \sum_i \sum_{i \sim j} \mathbb{P}(A_i \cap A_j)$. For a given (r, s) , the pairs of H_i, H_j that intersect r vertices and s edges contribute $\Theta(n^{2v-r} p^{2e-s}) = \Theta(\mu^2 / n^r p^s)$ to Δ . Note that

$$n^r p^s = (np^{s/r})^r > (np^{e/v})^r > n^{c'_r},$$

where $c'_r = r(1 - \alpha e/v) > 0$. Thus, there exists $c' > 0$, the smallest c'_r among all possible r , such that $\Delta = \Theta(\mu^2 / n^{c'})$. Thus $\mu^2 / 2(\mu + \Delta) = \Theta(n^{c''})$, where $c'' = \min(c, c')$. It follows from the third Janson's inequality that

$$\mathbb{P}(X = 0) \leq \exp(-\Theta(n^{c'})) \leq \exp(-n^\beta),$$

for $0 < \beta < c''$ and n large enough. □

Problem 9

Buffon's needle. A floor is made up of parallel strips of wood, each of width 1. A needle of length $\ell < 1$ is dropped onto the floor. What is the probability $p(\ell)$ that it crosses the boundary between two strips? (If the picture is not clear, Google for “Buffon's needle”!). The answer leads to an experimental method for estimating π .

The traditional solution is to average over the position of the end of the needle and over the angle at which the needle lies, using a double integral. This is reasonably straightforward, but an alternative approach is to use the principle of linearity of expectation.

Instead of $p(\ell)$, consider $e(\ell)$, the expected number of times that the needle crosses a line.

- (a) Show that if $\ell < 1$ then $p(\ell) = e(\ell)$.
- (b) Show that $e(\ell) = c\ell$ for some constant c (for all positive ℓ).
- (c) If we replace a straight needle by some other shape with the same length, the probability of crossing a line will change; however, note that the expectation of the number of line-crossings doesn't! So to calculate $e(\ell)$ we may consider “needles” which are not straight lines. Find a “needle” for which the expectation is very easy to calculate. Hence find c .

Proof.

□

Optional Problems: These questions are optional, but MFoCS students are recommended to attempt them. Outline solutions will be provided later; you can also approach class tutors (if they have time) or the lecturer to discuss them.

Problem 10

Let $X_1, \dots, X_n$ be independent random variables with $X_i \sim \text{Bernoulli}(p_i)$. Let $X := \sum_{i=1}^n X_i$ and let $Y \sim \text{Po}(\mu)$ where $\mu := \mathbb{E}[X] = \sum_{i=1}^n p_i$. Prove Le Cam's Theorem: for any $S \subseteq \mathbb{N}$,

$$|\mathbb{P}(X \in S) - \mathbb{P}(Y \in S)| \leq \sum_{i=1}^n p_i^2.$$

Hint: introduce $Y_i \sim \text{Po}(p_i)$ with $\mathbb{P}(X_i \neq Y_i) \leq p_i^2$.

Proof.

□

Problem 11

Let $p = \frac{\log n + c}{n}$ where c is constant and let X_n denote the number of isolated vertices in $G(n, p)$. Show that X_n tends in distribution to $\text{Po}(e^{-c})$ as $n \rightarrow \infty$.

Deduce that if $p = \frac{\log n + c}{n}$ where c is constant, then

$$\mathbb{P}(G(n, p) \text{ is connected}) \rightarrow e^{-e^{-c}}.$$

Proof.

□

Problem 12

Let $U \subseteq \mathcal{P}(X)$ be an up-set. The *influence* $I_p(i, U)$ of an element $i \in X$ on U is the probability that exactly one of $X_p \setminus \{i\}$ and $X_p \cup \{i\}$ is in U . Defining $f(p)$ as in Problem 5, show that

$$\frac{d}{dp} f(p) = \sum_{i \in X} I_p(i, U).$$

Proof.

□